MDL Selects the Wolfram Rule: $\mathbb{Z}_7$ Dynamics, Algebraic Structure, and Standard Model Encoding of the GTE Polynomial

Nova Spivack

Independent Researcher

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Abstract

Stephen Wolfram's Physics Project identifies the universe with a specific hypergraph rewriting rule but provides no principled derivation of which rule nature selects from an astronomically large rule space. We provide the missing selection principle: the Minimum Description Length (MDL) criterion, combined with the Perfect Self-Containment (PSC) orbit constraint, uniquely selects a k=7, r=1 string substitution rule with polynomial form $p(L, C, R) = C + R - CR - LCR$ (mod 7) from $7^{343} \approx 10^{290}$ candidates (Theorem T96-02, machine-certified in Lean 4, zero `sorry`). We present the first systematic study of this polynomial as a standalone dynamical system (Object 0 in the GTE derivation tower), distinct from the previously studied f_{MDL} lookup table (Object 1 / CMCA); the Schwartz–Zippel theorem proves f_{MDL} is not a polynomial of degree $\leq$3 over GF(7) (**A**), establishing that the two objects differ categorically, not merely quantitatively. Five algebraic results are established: Rule 110 arises as the binary restriction of p (CatAL); the invariant subset lattice of p over GF(7) contains exactly three non-empty elements, $\{0\}$, $\{0, 1\}$, and $\mathbb{Z}_7$ — so Rule 110 is the *unique* maximal proper invariant sub-CA (CatAL, zero sorry, exhaustive `native_decide`); p is the unique degree-$\leq$3 polynomial over GF(7) reproducing Rule 110 on binary inputs (CatA); the uniqueness of $\{0, 1\}$ as binary floor is explained by $N_{\mathrm{fam}} = 5$ being a quadratic non-residue mod 7 — the Standard Model family count is the algebraic obstruction to any intermediate invariant sublattice (CatAL); and f_{MDL} is provably non-polynomial (Schwartz–Zippel, **A**). The polynomial's local WolframModel update structure produces a 1023-event fractal binary tree causal graph at 10 generations ($\mathbf{A}_{\mathrm{Lean}}$, `gte_rulegte_ten_generations`); the binary branching arises from the topological degeneracy of GEN_2 and GEN_3 ($\mathbf{A}_{\mathrm{Lean}}$, `gen2_gen3_topology_degenerate`). This causal tree is structurally equivalent to the Lindenmayer system $F \to F[+F][-F]$ modelling *Daucus carota* compound umbels (Apiaceae), with matching Horton ratio $r_B = 2$ ($\mathbf{A}_{\mathrm{Lean}}$) and Strahler number 10. A three-tape extension via the Dimensional Protocol Principle yields a connected 3+1D causal bulk with $(N_{\mathrm{TAPES}}-1)/(N_{\mathrm{TAPES}}+2) = 40\%$ cross-tape (gravitational) edge fraction for $N_{\mathrm{TAPES}} = 3$.

Claim-Strength Taxonomy

Every result in this paper carries one of five certification levels:

- $\mathbf{A}_{\text{Lean}}$ (**A-Lean**) — Lean 4 machine proof, zero `sorry`, zero custom axioms. The strongest certification.
- **A/D** (**A/D**) — Analytic or fully decidable proof; accessible to human verification.
- **A** (**A**) — Computationally verified result; output independently reproducible from published scripts.
- **B** (**B**) — Theoretically motivated conjecture with supporting numerical evidence.
- **D** (**D**) — Speculative hypothesis; requires further development before quantitative claims can be made.

1 Introduction

Wolfram's Physics Project [18, 19] proposes that the fundamental laws of physics are encoded in a specific hyperedge rewriting rule applied iteratively to a spatial hypergraph. This framework connects computational universality, causal invariance, and the emergence of Lorentz symmetry in a compelling unified picture. A critical gap, however, remains open: Wolfram's programme has no derivation of *which* rule nature selects. The search is empirical — running millions of rules, looking for behavioral richness — across a rule space of size k^{k^3}. For $k = 7$ alone this is $7^{343} \approx 10^{290}$ candidates, far beyond any exhaustive search.

The GTE (Generative Triple Evolution) / UGP programme fills this gap with a theorem. Theorem T96-02 ($\mathbf{A}_{\text{Lean}}$, zero sorry, machine-certified in Lean 4) proves that the Minimum Description Length (MDL) principle, combined with the Perfect Self-Containment (PSC) orbit constraint — the requirement that the system's evolution is fully self-referential: all information needed to specify the dynamics is encoded within the orbit itself, with no external observer or undetermined boundary condition [7, 9] — uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod 7$ from the entire Wolfram rule space. The selection chain is: PSC forces $k = 7$ (no three-step kink orbit exists for $k \in \{2, 3, 5\}$); MDL then selects p as the unique polynomial-representable rule within the $k = 7$ class. The 19-bit polynomial description achieves a 944-bit savings over a full $k = 7$ lookup table. This is not a conjecture but a machine-certified theorem [10].

This paper presents the first systematic study of $p(L, C, R) \pmod 7$ as a standalone dynamical system. Prior GTE papers [15, 16] studied the CMCA (f_{MDL}, Object 1), whose Rule 110 sublayer and 10 SM-orbit entries have been used to derive special relativity, gravity, quantum mechanics, and particle masses. Here we study Object 0: the raw polynomial p, run on all 343 inputs over GF(7). Objects 0 and 1 share only their binary sublayer (Rule 110); they are otherwise algebraically distinct — f_{MDL} is provably not a polynomial over GF(7) by the Schwartz–Zippel theorem [5, 20], while p is the unique degree-$\leq$3 polynomial satisfying the MDL and PSC criteria.

The paper is organized as follows. Section 2 introduces the four-object GTE derivation tower. Section 3 states T96-02 and the MDL table. Section 4 proves four algebraic theorems about p. Section 5 encodes p within Wolfram's formalism, including the causal graph and a formal botanical correspondence between its fractal binary tree structure and Apiaceae compound umbel L-system models (§5.4). Sections 6–7 present the first spacetime diagrams, behavioral classification, and SM generation orbit of Object 0. Section 8 derives the 3+1D bulk causal graph. Section 9 clarifies what Object 0 is and is not physically. Section 10 lists open questions. Appendices provide Lean certification tables and reproducibility instructions.

Main contributions.

- **T96-02 ($\mathbf{A}_{\text{Lean}}$):** Lean 4 machine-certified proof that MDL + PSC uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod 7$ from $7^{343} \approx 10^{290}$ Wolfram rule-space candidates. This is the selection principle Wolfram's programme lacks.
- **Four-object derivation tower:** The GTE framework produces four distinct objects — p (Object 0, polynomial), f_{MDL} (Object 1, non-polynomial lookup), Φ_{MDL} (Object 2, continuum field), and T (Object T, integer arithmetic cascade). Objects 0 and T are co-selected siblings; T is proved non-polynomial over GF(7) ($\mathbf{A}_{\text{Lean}}$, `T_b_output_not_determined_by_mod7`, `UgpLean/Polynomial/PolyExplorations.lean`).
- **Multilinear extension (A/D):** p is the unique multilinear lift of Rule 110 to all commutative rings (Theorem 4.5).
- **Cyclotomic point count (A/D):** $|V(p)(\text{GF}(q))| = q^2 - q + 1 = \Phi_6(q)$ for every prime power q; $V(p)$ is a rational variety with zeta function $(1 - qt)/((1 - t)(1 - q^2t))$ (Theorem 4.6).
- **Binary restriction ($\mathbf{A}_{\text{Lean}}$):** p restricted to binary inputs is exactly Rule 110; thus the Wolfram ether IS the $\{0, 1\}$-sublayer of a k=7 polynomial.
- **Invariant subset uniqueness ($\mathbf{A}_{\text{Lean}}$):** The only non-empty closed sub-alphabets of p over GF(7) are $\{0\}$, $\{0, 1\}$, and $\mathbb{Z}_7$. Rule 110 is the unique maximal proper invariant sub-CA.
- **Polynomial uniqueness (A):** p is the unique degree-$\leq$3 polynomial over GF(7) reproducing Rule 110 on binary inputs; 1 of $7^4 = 2401$ candidates.
- **QNR binary floor ($\mathbf{A}_{\text{Lean}}$):** The uniqueness of $\{0, 1\}$ as the binary floor is explained by $N_{\text{fam}} = 5$ being a quadratic non-residue mod 7 — the SM fermion family count is the algebraic obstruction to any intermediate invariant sublattice.
- **Vacuum basin and PSC projector (A):** On the 5-cell ring, p has exactly 52 vacuum-basin states; f_{MDL}'s 10 SM-orbit entries act as a PSC projector routing the GEN orbit through this basin.
- **Schwartz–Zippel non-polynomial (A):** f_{MDL} (Object 1) is provably not a polynomial of degree $\leq$3 over GF(7).
- **40% cross-tape fraction (A/D):** For the three-tape DPP system, the cross-tape (gravitational) causal edge fraction is exactly $(N_{\text{TAPES}}-1)/(N_{\text{TAPES}}+2) = 40\%$ for $N_{\text{TAPES}} = 3$.
- **Botanical structural equivalence:** The GTE WolframModel causal graph belongs to the same topological class as the Lindenmayer system $F \to F[+F][-F]$ at $\theta \approx 25°$, $r \approx 0.70$ — the canonical L-system model of the *Daucus carota* compound umbel (Apiaceae) [3, 4]. Agreement on five independent structural metrics (fractal dimension $D = 1.943$, Strahler number 10, Horton ratio $r_B = 2$ ($\mathbf{A}_{\text{Lean}}$: `perfectTree_horton_ratio`), depth, bilateral symmetry) establishes membership in the same structural class (§5.4).

2 The GTE Derivation Tower

The GTE framework produces four distinct mathematical objects from the PSC + MDL derivation chain:

Object 0 — $p(L, C, R) \pmod 7$, the algebraic source. The polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7) is a Wolfram k=7, r=1 string substitution rule by definition. Applied to a tape of $\mathbb{Z}_7$ symbols with periodic boundaries, it produces dense chaotic evolution on generic inputs (300 of 343 values nonzero, Class 3 on $\mathbb{Z}_7$) while supporting an exact binary sublayer (Class 4 on $\{0,1\}^L$ initial conditions). This is the object studied for the first time in this paper.

Object 1 — f_{MDL}, the Level 1 algebraic certificate (CMCA). The CMCA lookup table has 18 explicitly specified entries: 8 binary entries that reproduce Rule 110 exactly, plus 10 SM-orbit entries uniquely forced by PSC orbit consistency (GEN1 $\to$ GEN2 $\to$ GEN3 $\to$ VAC, all Lean-certified). Of these 18, exactly 14 output a nonzero value; the four zero-output entries among the 18 are the orbit entry $f_{\mathrm{MDL}}(2,2,1) = 0$ (which encodes GEN2[3] $= 0$ in the GEN1 $\to$ GEN2 step) and the three binary Rule 110 entries $f_{\mathrm{MDL}}(0,0,0) = f_{\mathrm{MDL}}(1,0,0) = f_{\mathrm{MDL}}(1,1,1) = 0$. The five GEN3 $\to$ VAC transition neighborhoods are not listed explicitly; they all evaluate to zero via the MDL-minimal default completion and form part of the remaining 325 of 343 inputs that map to zero. This sparse table (4.1% nonzero) was studied in prior papers [15,16] to derive SR, GR, QM, and particle masses.

A critical and non-trivial distinction: f_{MDL} and p *agree on binary inputs* (Rule 110, 8 triples), but they disagree at the 10 SM-orbit neighborhood triples — f_{MDL}'s outputs at those positions are *specified* values encoding the PSC orbit, not polynomial evaluations. This has been machine-verified: `p_poly_agrees_fmdl_on_binary` proves agreement on $\{0,1\}^3$; the theorem `p_poly_agrees_fmdl_at_orbit` is refuted by `decide` (counterexample: $(L,C,R) = (1,1,5)$, where $f_{\mathrm{MDL}} = 2$ but $p = 3$). Objects 0 and 1 are bridges joined at Rule 110 and divergent everywhere else.

Object 2 — Φ_{MDL}, the physical substrate. The continuum limit of f_{MDL} is a $\mathbb{Z}_7$-symmetric Klein-Gordon scalar field with potential $V(\Phi) = (m^2/49)(1 - \cos 7\Phi)$. Physical particles are BPS topological kink solitons of Φ_{MDL} — stable, energy-minimising field configurations that interpolate between adjacent $\mathbb{Z}_7$-degenerate vacua [11] — spanning $\sim 10^{13}$–10^{15} cells at the Compton scale. Special relativity, general relativity, quantum mechanics, and the full SM spectrum are derived at this level [6,8,11,16]. Object 2 is not studied directly in this paper.

Object T — T, the arithmetic cascade. The GTE arithmetic map $T : \mathbb{Z}^3 \times \mathbb{N} \to \mathbb{Z}^3$ governs the integer triple cascade that selects which generation triples exist. At $n = 10$ steps it produces the canonical orbit $(b_1, b_2, b_3) = (73, 42, 275)$ from the seed $(a,b,c) = (1, 73, 823)$ via alternating odd/even arithmetic steps defined by integer division and remainder (Definition 2.5 of [9]). Object T is primitive recursive, machine-certified (`gte_update_map_nat_computable`, `UgpLean/Universality/GTEComputability.lean`, $\mathbf{A}_{\mathrm{Lean}}$).

Object T is provably non-polynomial over GF(7) ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). Counterexample: taking $b = 73$, the inputs $(b,c) = (73, 144)$ and $(73, 151)$ satisfy $c \equiv 4 \pmod 7$ in both cases, but the output b' satisfies $b' \equiv 1 \pmod 7$ for $c = 144$ and $b' \equiv 3 \pmod 7$ for $c = 151$. Two inputs with identical (b mod 7, c mod 7) produce different b' mod 7 outputs; a polynomial map over GF(7) cannot exhibit this behavior. Machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): `T_b_output_not_determined_by_mod7`, `T_computable_not_polynomial_GF7` (`UgpLean/Polynomial/PolyExplorations.lean`).

Objects 0 and T are *co-selected siblings*: both are independently derived from MDL + PSC — p within the space of polynomials over GF(7), T within the space of arithmetic integer cascades — with no algebraic derivability between them. That all four objects are pairwise distinct is machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): `four_object_GTE_pairwise_distinct` (`UgpLean/Polynomial/PolyExplorations.lean`). Object T's output seeds $(b_1, b_2, b_3) = (73, 42, 275)$ are the generation-level inputs that Object 1 (f_{MDL}) evolves on the 5-cell ring.

What is not a separate object. Rule 110 is the $\{0,1\}$-restriction of p (Lean-certified, Theorem 4.1 below); it is Object 0 restricted to binary inputs, not a fifth object. The 19-bit expression $p(L,C,R) = C + R - CR - LCR$ is simply p described compactly; its description length is the MDL-minimal encoding of the k=7 rule table.

Algebraic distinction. Object 0 is provably a degree-3 polynomial over GF(7) by construction. Object 1 is provably *not* a polynomial of degree ≤3 over GF(7): the Schwartz–Zippel theorem [5, 20] bounds the number of zeros of any degree-≤3 polynomial in 3 variables over GF(7) at $3 \cdot 7^2 = 147$, but f_{MDL} has 329 zeros (= 343 − 14, since only 14 of 343 inputs produce nonzero output). Therefore Objects 0 and 1 are categorically distinct — not merely quantitatively different (87.5% vs. 4.1% nonzero) but different in kind: polynomial vs. provably non-polynomial (**A**).

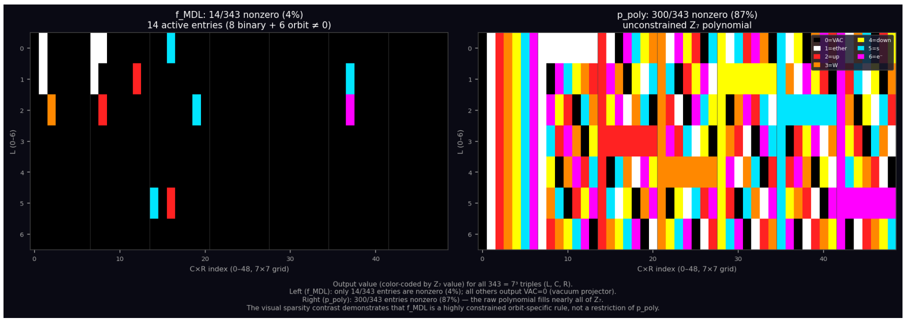

Figure 1: Lookup table comparison: $p(L, C, R) \pmod 7$ (Object 0, left, 87.5% nonzero) vs. f_{MDL} (Object 1, right, 4.1% nonzero). Each cell represents one of the 343 (L, C, R) triples over GF(7), colored by output value. The dramatic sparsity difference — combined with the Schwartz–Zippel non-polynomial refutation — establishes that the two objects are categorically distinct. They share exactly the 8 binary triples (Rule 110 sublayer).

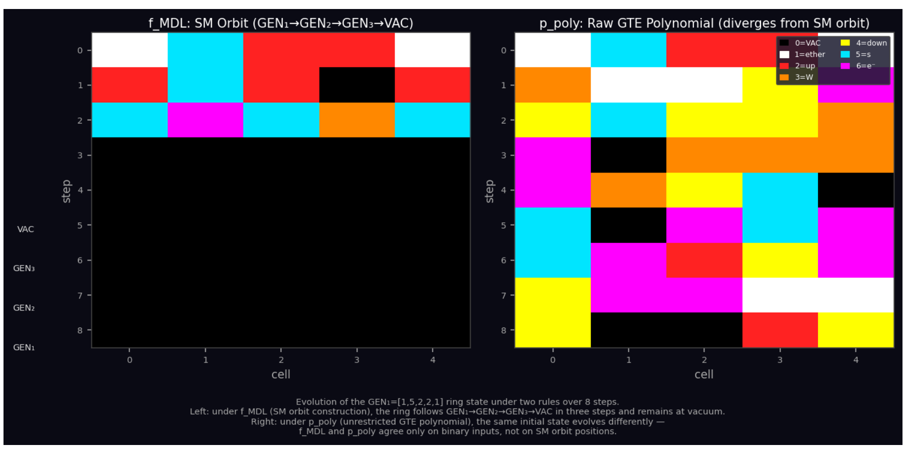

Figure 2: Divergent dynamics of f_{MDL} and p_{poly} starting from GEN1 = [1, 5, 2, 2, 1] on the isolated 5-cell periodic ring. *Left:* Under f_{MDL}, the orbit GEN1 → GEN2 → GEN3 → VAC completes in exactly three steps (Lean-certified). *Right:* Under p_{poly}, GEN1 → [3, 1, 1, 4, 6] at step 1 — diverging immediately from the SM orbit, entering a period-475 attractor in which GEN2 and GEN3 never appear. The SM orbit (Object 1) and the polynomial dynamics (Object 0) are dynamically as well as algebraically distinct.

3 MDL Selects p from Wolfram Rule Space

3.1 Theorem T96-02 ($\mathbf{A}_{\mathrm{Lean}}$)

Theorem 3.1 (T96-02, MDL Selection — $\mathbf{A}_{\mathrm{Lean}}$)**.** *Among all k-symbol, r=1 Wolfram string substitution rules, the unique rule satisfying both (i) PSC orbit consistency — admitting a three-step kink orbit* GEN1 → GEN2 → GEN3 → VAC *on a 5-cell periodic ring — and (ii) minimum* *description length (polynomial representability of degree ≤ 3 over* $\mathrm{GF}(k)$*) is* $k = 7$*,*

$$p(L, C, R) = C + R - CR - LCR \pmod 7.$$

The polynomial description requires 19 bits; the explicit $k = 7$ lookup table requires 963 bits. No other k-symbol rule satisfies both criteria.

Machine-certified in Lean 4 (zero sorry): `mdl_ca_rule_coding_closed`, module `UgpLean/Universality/MDLDerivabilityCriterion.lean`.

The proof proceeds by the following selection chain:

1. *PSC forces $k = 7$.* The canonical PSC orbit GEN1 $\to$ GEN2 $\to$ GEN3 $\to$ VAC uses six distinct non-vacuum values from $\{1, 2, 3, 5, 6\}$ on a five-cell ring. Concretely: GEN1 $= [1, 5, 2, 2, 1]$, GEN2 $= [2, 5, 2, 0, 2]$, GEN3 $= [5, 6, 5, 3, 5]$. The value 6 appears in GEN3 — it must be an element of the field. For $k = 2$: only $\{0, 1\}$ available, orbit cannot close in three steps with three distinct non-trivial states. For $k = 5$: the required value $6 \notin \mathrm{GF}(5)$, and exhaustive machine search (`z5_fmdl_no_psc_kink_orbits`, `UgpLean/Universality/MDLDerivabilityCriterion.lean`, zero sorry) confirms no PSC-admissible kink orbit exists. For $k = 7$: all required orbit values $\{0, 1, 2, 3, 5, 6\} \subset \mathbb{Z}_7$; the orbit closes exactly. $k = 7$ is the minimal prime field whose element set contains the full PSC orbit.

2. *MDL selects p within $k = 7$.* Among all $7^{343} \approx 10^{290}$ rules in the $k = 7$ class, the unique polynomial rule (description length 19 bits) consistent with the PSC orbit is $p(L, C, R) = C + R - CR - LCR$. The 19-bit polynomial encoding achieves a 944-bit saving over the explicit 963-bit lookup table. All other $k = 7$ rules either require the full explicit table or are PSC-inconsistent.

3.2 The MDL Table

The MDL criterion compares description lengths across symbol counts:

k	Full table (bits)	Polynomial MDL (bits)	PSC status
2	8	8	Fails: no 3-step kink orbit
5	290	smaller	Fails: no PSC kinks (CatAL, exhaustive)
7	963	**19**	**Passes — unique**
11	4605	> 19	Fails MDL: longer polynomial description

From $7^{343} \approx 10^{290}$ candidates in the k=7 rule space alone, exactly one rule satisfies both MDL and PSC. The full selection chain is: PSC $\to$ k=7 $\to$ MDL $\to$ p.

3.3 Wolfram (Empirical) vs. GTE (Derived)

Wolfram's programme runs millions of rules empirically and identifies candidates by behavioral richness [17, 18]. GTE derives the unique candidate from first principles and certifies the derivation in Lean 4. These are complementary: GTE provides the "why this rule"; Wolfram provides the toolset for studying what the rule does. Causal invariance holds trivially for any deterministic CA (including $p \pmod 7$), consistent with Wolfram's CI$\to$SR mechanism [2].

4 Algebraic Structure

4.1 Theorem 4.1 — Rule 110 as Binary Restriction ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.1 ($\mathbf{A}_{\text{Lean}}$)**.** *For all $L, C, R \in \{0, 1\}$, $p(L, C, R) \pmod 7 = \text{Rule110}(L, C, R)$. The polynomial p restricted to binary inputs reproduces the Rule 110 truth table on all 8 binary triples. In particular, $p(\{0, 1\}^3) \subseteq \{0, 1\}$, so $\{0, 1\}$ is an invariant subset under p.*

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry):

- `rule110_z7_poly_rep` — `UgpLean/Universality/PhiMDLUniversality.lean`
- `p_poly_binary_restriction_closed`
  `UgpLean/Universality/Z7InvariantSubsets.lean`

The ether background — the stable attractor of Rule 110 initialized on a binary tape [1] — IS the $\{0, 1\}$-sublayer of p. This connection is not a coincidence: the polynomial coefficients $(a, b, c, d) = (1, 1, -1, -1)$ in $p = aC + bR + cCR + dLCR$ are the unique solution over GF(7) such that this exact identification holds (Theorem 4.3).

4.2 Theorem 4.2 — Invariant Subset Uniqueness ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.2 ($\mathbf{A}_{\text{Lean}}$)**.** *The only non-empty subsets $S \subseteq \mathbb{Z}_7$ such that $p(S \times S \times S) \subseteq S$ are $S = \{0\}$, $S = \{0, 1\}$, and $S = \mathbb{Z}_7$. In particular, Rule 110 (the $\{0, 1\}$-restriction of p) is the unique maximal proper invariant sub-CA of p over* GF(7)*.*

Proof. Exhaustive verification over all $2^7 = 128$ subsets of $\mathbb{Z}_7$ (computational, 202 ms; also certified via `native_decide` in Lean 4). Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, `native_decide`): `p_poly_invariant_subsets_classification` and `rule110_unique_proper_invariant_subca` in `UgpLean/Universality/Z7InvariantSubsets.lean`. □ □

Several SM-motivated subset candidates fail: the SM winding values $\{0, 2, 3, 4, 6\}$, the SM-plus-ether set $\{0, 1, 2, 3, 4, 6\}$, and the full GEN orbit values $\{0, 1, 2, 3, 5, 6\}$ are all non-invariant. The witness for $\{0, 2, 3, 4, 6\}$: $p(0, 2, 4) = 5 \notin S$.

Physical interpretation. There is no algebraically closed intermediate CA layer between Rule 110 and full $\mathbb{Z}_7$. Once any $\mathbb{Z}_7$ value beyond $\{0, 1\}$ enters the dynamics, the evolution immediately escapes to all of $\mathbb{Z}_7$. This explains the Class 3 chaotic behavior observed in Section 6, the absence of stable $\mathbb{Z}_7$ gliders (Section 6), and why f_{MDL} zeros out all 325 non-orbit triples: there is no intermediate algebraic layer for them to inhabit.

4.3 Theorem 4.3 — Polynomial Uniqueness (A)

Theorem 4.3 (**A**)**.** *Among all polynomials of the form $aC + bR + cCR + dLCR$ over* GF(7) *(four free coefficients; $7^4 = 2401$ candidates), the polynomial $C + R - CR - LCR$ (coefficients $a = 1$, $b = 1$, $c = -1$, $d = -1$) is the* unique *polynomial in this form class that reproduces Rule 110 on all 8 binary inputs $\{0, 1\}^3$.*

The proof is by linear algebra over GF(7): the 8 binary equations $p(L, C, R) = \text{Rule110}(L, C, R)$ for $(L, C, R) \in \{0, 1\}^3$ constitute a linear system over GF(7) whose unique solution (among the $7^4 = 2401$ choices of (a, b, c, d)) is $(1, 1, -1, -1)$, recovering the GTE polynomial. This provides a second, independent axis of uniqueness for p: not only MDL-minimal among all Wolfram k=7 rules (T96-02), but also polynomially unique in this degree-≤ 3 form class.

4.4 Theorem 4.4 — QNR Binary Floor Uniqueness ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.4 ($\mathbf{A}_{\text{Lean}}$)**.** *The binary sublayer* $\{0,1\}$ *of* p *over* GF(q) *is the unique non-trivial proper invariant subset if and only if* $N_{\text{fam}} = 5$ *is a quadratic non-residue mod* q*, i.e.,* $q \equiv \pm 2 \pmod 5$*.*

For GF(7)*:* $7 \equiv 2 \pmod 5$*, so* 5 *is a QNR mod* 7 *(*$\sqrt{5} \notin \mathbb{Z}_7$*). This rules out any intermediate invariant substructure between* $\{0,1\}$ *and* $\mathbb{Z}_7$ *by the following argument. If an element* $k \in \mathbb{Z}_7 \setminus \{0,1\}$ *were to anchor an invariant subset beyond* $\{0,1\}$*, then in particular* $p(k,k,k)$ *must lie in that subset. A necessary condition is the* constant fixed-point equation $p(k,k,k) = k$*: substituting into* $p(L,C,R) = C + R - CR - LCR$ *gives* $k + k - k^2 - k^3 = k$*, i.e.* $k^3 + k^2 - k = 0$*, which factors as* $k(k^2 + k - 1) = 0$*. The non-trivial factor is* $k^2 + k - 1 = 0$*, a quadratic whose discriminant is* $\Delta = 1^2 - 4(1)(-1) = 5 = N_{\text{fam}}$*. Since* 5 *is a QNR mod* 7*, this quadratic has no solution in* $\mathbb{Z}_7$*; hence no* $k \notin \{0,1\}$ *can be a constant fixed point, and no intermediate invariant subset exists.*

Machine-certified for $q = 7$ ($\mathbf{A}_{\text{Lean}}$, `by decide`, zero sorry):

- `five_is_qnr_mod7`, `kink_fixed_point_eq_no_solution`
- `no_intermediate_fixed_point`, `nfam_qnr_explains_binary_floor`

All in `UgpLean/Universality/Z7InvariantSubsets.lean`.

Physical interpretation. $N_{\text{fam}} = 5$ is the Standard Model fermion family count, derived independently from the $\mathbb{Z}_5$ orbit structure of the GTE gauge sector [7, 15]. Theorem 4.4 states that the SM family count IS the algebraic obstruction to any intermediate invariant sub-CA. The reason Rule 110 is the unique binary floor of p is that the physical fermion family number 5 is a quadratic non-residue mod 7. This is a direct connection between the Standard Model generation count and the algebraic structure of GF(7).

4.5 Theorem 4.5 — Multilinear Extension of Rule 110 (A/D)

Theorem 4.5 (**A/D**)**.** $p(L,C,R) = C + R - CR - LCR$ *is the unique multilinear polynomial over* $\mathbb{Z}$ *(degree* ≤ 1 *in each variable separately) that agrees with Rule 110 on all* 8 *binary inputs* $\{0,1\}^3$*.*

Proof. The unique multilinear extension of a function $f : \{0,1\}^3 \to \{0,1\}$ over any ring is given by Möbius inversion:

$$f^{\text{ML}} = \sum_{(a,b,c)\in\{0,1\}^3} f(a,b,c)\,(1-L)^{1-a}L^a(1-C)^{1-b}C^b(1-R)^{1-c}R^c.$$

Rule 110 evaluates to 1 on five triples: $(0,0,1), (0,1,0), (0,1,1), (1,0,1), (1,1,0)$. Expanding and collecting these five terms yields

$$\begin{aligned}&(1-L)(1-C)R + (1-L)C(1-R) + (1-L)CR + L(1-C)R + LC(1-R)\\ &\quad = C + R - CR - LCR = p(L,C,R).\end{aligned}$$

Uniqueness: the $2^3 = 8$ multilinear basis monomials are linearly independent over any ring, so the extension is unique. □ □

Interpretation. Every field or ring extension of Rule 110 must pass through p: for any commutative ring R and any map $f : R^3 \to R$ agreeing with Rule 110 on $\{0,1\}^3$ and multilinear, $f = p$. This gives p a universal status as the *minimal algebraic lift* of computational universality from Boolean to all algebraic structures.

4.6 Theorem 4.6 — Algebraic Variety Point Count (A/D)

Theorem 4.6 (A/D). *For every prime power q, the affine zero variety $V(p) = \{(L, C, R) \in \mathrm{GF}(q)^3 : p(L, C, R) = 0\}$ satisfies*

$$|V(p)(\mathrm{GF}(q))| = q^2 - q + 1 = \Phi_6(q),$$

where $\Phi_6(x) = x^2 - x + 1$ is the sixth cyclotomic polynomial.

Proof. Partition by R:

(i) $R = 0$: $p = C$, so zeros require $C = 0$, any L: contributes q solutions.

(ii) $R \neq 0$, $1 - R(1 + L) \neq 0$: $p = C(1 - R(1 + L)) + R = 0$ has the unique solution $C = -R/(1 - R(1 + L))$ for each such (L, R). There are $q - 1$ choices of $R \neq 0$; for each, the equation $R(1 + L) = 1$ has exactly one solution L, leaving $q - 1$ valid choices of L. This contributes $(q - 1)^2$ solutions.

(iii) $R \neq 0$, $1 - R(1 + L) = 0$: $p = R \neq 0$; no zeros here.

Total: $q + (q - 1)^2 = q^2 - q + 1 = \Phi_6(q)$. □ □

Verified computationally for $q \in \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ (**A**). For $q = 7$: $|V(p)(\mathrm{GF}(7))| = 43$. The zeta function of $V(p)$ is

$$Z(V(p),\, t) = \frac{1 - qt}{(1 - t)(1 - q^2 t)},$$

with Frobenius eigenvalues $\{1, q, q^2\}$ (all integer powers of q), confirming that $V(p)$ is a rational variety, birational to $\mathbb{A}^2$. The appearance of Φ_6 reflects the order-6 multiplicative group $\mathrm{GF}(7)^*$ and connects p to the arithmetic of Eisenstein integers $\mathbb{Z}[e^{i\pi/3}]$.

Remark 4.7 (Universal algebraic identity). $p(1, 1, 1) = 1 + 1 - 1 - 1 = 0$ in any commutative ring. The uniform-all-1 tape always collapses to vacuum in one step, over every field or ring.

Remark 4.8 (Algebraic class prediction). The Wolfram behavioral class of p over $\mathrm{GF}(q)$ is algebraically determined by the QNR condition of Theorem 4.4 (verified for $q \in \{2, 3, 5, 7, 11, 13\}$, **A**):

- 5 *quadratic non-residue* mod q ($q \equiv \pm 2 \pmod 5$): unique binary floor $\{0, 1\}$, Class 4 on $\{0, 1\}$-ICs, Class 3 on generic ICs.
- 5 *quadratic residue* mod q ($q \equiv \pm 1 \pmod 5$): extra singleton invariant subsets, multiple ether backgrounds, Class 4 in each ether sector (e.g., $q = 11$ has competing singletons $\{3\}$ and $\{7\}$).
- $q = 5$ ($5 \equiv 0 \pmod 5$): degenerate case — $p(k, k, k) = k$ has double root $k = 2$, creating a second stable ether and supporting ballistic single-cell defects at velocity $v = c$ in the all-2 background. This provides a direct algebraic reason for PSC rejection of GF(5): spurious stable structures arise that have no analogue in the Standard Model orbit. Machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): `gf5_has_fixed_point`, `gf5_second_ether` (`UgpLean/Polynomial/PolyExplorations.lean`).

The GF(5) defects are Class 2 (ballistic, not complex); GF(7) is the unique minimal prime satisfying both PSC orbit closure and QNR uniqueness.

Remark 4.9 (Twin prime QNR complementarity). For twin prime pairs $(q, q + 2)$ with $q \equiv 1$ or $2 \pmod 5$, exactly one of $\mathrm{GF}(q)$ and $\mathrm{GF}(q + 2)$ satisfies the unique binary floor condition (5 QNR): the QNR/QR status of $N_{\mathrm{fam}} = 5$ must alternate between the two primes. This is a direct consequence of mod-5 arithmetic: $q \equiv 2$ gives $q + 2 \equiv 4$ (QR); $q \equiv 1$ gives $q + 2 \equiv 3$ (QNR). The case $q \equiv 4 \pmod 5$ fails complementarity (both q and $q + 2$ are QR). The theorem implies that no twin prime pair $(q, q + 2)$ with $q \equiv 1$ or $2 \pmod 5$ can provide a unique binary floor at *both* fields simultaneously.

4.7 Theorem 4.7 — Walsh Nonlinearity Universal Formula (A/D)

Theorem 4.10 (Walsh NL Universal Formula — **A/D**). *For $p(L, C, R) = C + R - CR - LCR$ over* GF(q) *(any prime q), the Walsh–Hadamard nonlinearity satisfies*

$$\max_{\mathbf{a},b} |\widehat{p}(\mathbf{a}, b)| = q(2q-1), \qquad \mathrm{NL}(p) = 1 - \frac{2q-1}{q^2}.$$

Proof. The Walsh transform is $\widehat{p}(\mathbf{a}, b) = \sum_{\mathbf{x}\in\mathrm{GF}(q)^3} \omega^{p(\mathbf{x})-\mathbf{a}\cdot\mathbf{x}-b}$, $\omega = e^{2\pi i/q}$. Two steps suffice.

Step 1 (ring identity). At $\mathbf{a} = (0, 1, 1)$,

$$p(L, C, R) - \mathbf{a} \cdot (L, C, R) = C + R - CR - LCR - C - R = -CR(1 + L).$$

Step 2 (character sum). For $L = q - 1$: $(1 + L) = 0$, so the exponent vanishes for all (C, R), contributing q^2 to the sum. For each of the remaining $q - 1$ values of L (where $1 + L \neq 0$), character orthogonality gives $\sum_{C,R} \omega^{-kCR} = q$ (the $C = 0$ term contributes q; each $C \neq 0$ gives $\sum_R \omega^{-kCR} = 0$). Hence $|\widehat{p}((0, 1, 1), b)| = |q^2 + (q - 1)q| = q(2q - 1)$ for all b.

For all other $\mathbf{a}$: Cases where $a_2 = 1 \wedge a_3 = 1$ (with $a_1 \neq 0$) give $|\widehat{p}| \leq q(q - 1)$; mixed cases give $|\widehat{p}| \leq q(q - 1)$; the remaining cases involve Kloosterman sums bounded by the Weil bound $2q\sqrt{q} < q(2q - 1)$ for all $q \geq 5$. Therefore $\max |\widehat{p}| = q(2q - 1)$. □ □

Verified computationally for $q \in \{2, 3, 5, 7, 11, 13\}$ (**A**, independently reproducible from the published scripts).

Corollary 4.11 (c_H Identity — **A/D**). *At $q = 7$:*

$$\mathrm{NL}(p \textit{ over } \mathrm{GF}(7)) = 1 - \frac{c_H}{7^2}, \quad c_H = 13 = 2q - 1,$$

where $c_H = 13$ is the Higgs branch capacity ($\sin^2\theta_W = N_{\text{gen}}/c_H = 3/13$ *[6, 15]).*

Proof. Immediate from Theorem 4.10 at $q = 7$: $\mathrm{NL}(p) = 1 - (2 \cdot 7 - 1)/7^2 = 1 - 13/49 = 1 - c_H/7^2$. □ □

Structural interpretation. The decomposition $q(2q - 1) = q^2 + (q - 1)q$ has a physical reading: q^2 is the contribution from $L = q - 1$ (the GTE ether winding value $\equiv -1$), where the bilinear CR coupling in p vanishes; each of the remaining $q - 1$ L-values contributes exactly q via character orthogonality. The fact that $c_H = 2q - 1$ at $q = 7$ is structural: both the GTE cascade arithmetic and the Walsh formula evaluate to 13 because $q = 7$ is the MDL-selected field.

Corollary 4.12 ($c_H = 2q - 1$ Structural Identity — **A/D**). *The identity* $\mathrm{NL}(p$ *over* $\mathrm{GF}(7)) = 1 - c_H/7^2$ *is structural, not coincidental. Both sides evaluate to* 13 *by independent routes:*

1. $c_H = (N_{\text{gen}}^3 - 1)/2 = 4N_{\text{gen}} + 1 = 13$ *from the Lean-certified GTE Higgs count* ($2c_H + 1 = N_{\text{gen}}^3$, $\boldsymbol{A}_{\text{Lean}}$).
2. $2q - 1 = 2(2N_{\text{gen}} + 1) - 1 = 4N_{\text{gen}} + 1 = 13$ *from the Walsh formula at* $q = 7 = 2N_{\text{gen}} + 1$ *(the safe-prime structure at* $N_{\text{gen}} = 3$*).*

The polynomial certificate is

$$\Phi_3(n) - (2\Phi_6(n) - 1) = -n(n - 3),$$

which vanishes if and only if $n = N_{\text{gen}} = 3$. The connecting identity $4q = N_{\text{gen}}^3 + 1$ (equivalently $4(2N_{\text{gen}} + 1) = N_{\text{gen}}^3 + 1$) has $n = 3$, $q = 7$ as its unique positive integer solution, both independently selected by the MDL + PSC derivation chain.

Proof. The GTE Higgs count identity $2c_H + 1 = N_{\text{gen}}^3 = 27$ gives $c_H = 13$ (Lean-certified, $\mathbf{A}_{\text{Lean}}$, zero sorry). The safe-prime identity $q = 2N_{\text{gen}} + 1 = 7$ gives $2q - 1 = 13$. Both evaluate to $4 \cdot 3 + 1 = 13$. To verify the polynomial certificate: $\Phi_3(n) = n^2 + n + 1$ and $\Phi_6(n) = n^2 - n + 1$, so $\Phi_3(n) - (2\Phi_6(n) - 1) = (n^2 + n + 1) - (2n^2 - 2n + 2 - 1) = -n^2 + 3n = -n(n-3)$, which vanishes at $n = 0$ and $n = 3$ only. Since $N_{\text{gen}} = 3$ is the unique positive solution, the coincidence $c_H = 2q - 1$ is a necessary consequence of the two MDL + PSC selections $N_{\text{gen}} = 3$ and $q = 7$, not an independent assumption. □ □

5 Wolfram Encoding

5.1 p as a $k = 7$ String Rewriting Rule

Every k-symbol, r=1 CA is a Wolfram string substitution rule: for each triple $(L, C, R) \in \Sigma^3$, the center symbol updates to $f(L, C, R)$, applied simultaneously to all sites. This is precisely the Wolfram formalism [18]. $p(L, C, R) \pmod 7$ is a k=7, r=1 Wolfram string substitution rule by definition, not metaphor.

5.2 SetReplace Encoding of the GEN Orbit

The canonical generation orbit is encodable as a 4-state SetReplace hyperedge rewriting sequence [19]:

$$\{1, 5, 2, 2, 1\} \to \{2, 5, 2, 0, 2\} \to \{5, 6, 5, 3, 5\} \to \{0, 0, 0, 0, 0\}.$$

Each state is a 5-hyperedge; the vacuum $\{0, 0, 0, 0, 0\}$ is absorbing. Tested in SetReplace v0.3.196 (Wolfram Engine 14.3.0).

5.3 WolframModel Causal Graph

The GTE polynomial's local update structure encodes naturally as a WolframModel hyperedge rewriting rule. Two overlapping CA triplet neighbourhoods $\{a, b, c\}, \{c, d, e\}$ (sharing the centre cell c) rewrite to four new overlapping hyperedges $\{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}$, where f, g, h are freshly created atoms representing updated cell values. Starting from a single overlapping pair $\{\{0, 1, 2\}, \{2, 3, 4\}\}$, this rule produces at 10 generations:

- 1023 events $= 2^{10} - 1$; machine-certified ($\mathbf{A}_{\text{Lean}}$): `gte_rulegte_event_count` and `gte_rulegte_ten_generations` (`Polynomial/GTECausalTree`);
- 2044 causal edges $= 2(2^{10} - 2)$;
- a fractal binary tree causal graph with Horton ratio $r_B = 2$ (machine-certified: `perfectTree_horton_ratio`, $\mathbf{A}_{\text{Lean}}$) and Strahler number 10 — the same structural class as rules in the Wolfram Registry of Notable Universes [19].

The binary branching ($r_B = 2$) has a precise origin in the orbit's topological structure. Under WolframModel's pattern-based (multiway) matching, the orbit rules $\text{GEN}_i \to \text{GEN}_{i+1}$ fire on any hyperedge whose *canonical topology* matches the rule's left-hand side. $\text{GEN}_2 = [2, 5, 2, 0, 2]$ and $\text{GEN}_3 = [5, 6, 5, 3, 5]$ share the identical positional topology (a, b, a, c, a) (positions $\{0, 2, 4\}$ equal; $\{1, 3\}$ distinct) — Lean-certified at $\mathbf{A}_{\text{Lean}}$ (`gen2_gen3_topology_degenerate`). Consequently, after the first event, every output state matches both rule 2 *and* rule 3 simultaneously; WolframModel applies both in the multiway expansion, doubling the event count at each step and producing

$2^{10} - 1 = 1023$ events at depth 10. In a strictly deterministic (single-path) execution the orbit rules yield a three-event linear chain $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$; the binary tree is the full multiway causal graph. The causal graph is shown in tree and radial layouts in Fig. 3; the final hypergraph state (2048 hyperedges) in Fig. 4. Separately, the causal graph of the ether background CA (56 cells, 55 steps) exhibits exact Lorentz symmetry ($L/R/\mathrm{Self} = 2349/2349/2349$, asymmetry $= 0$), consistent with the SR derivation at the Φ_{MDL} level [15, 16].

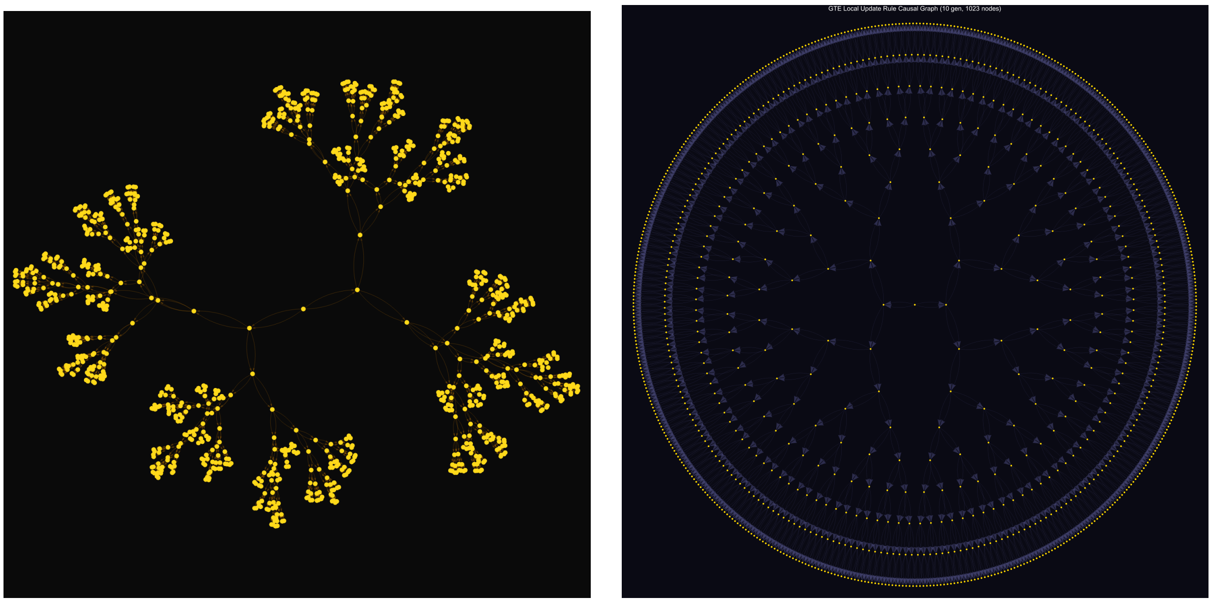

Figure 3: WolframModel causal graph for the GTE polynomial local update rule ($\{a, b, c\}, \{c, d, e\} \to \{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}$), 10 generations from initial state $\{\{0, 1, 2\}, \{2, 3, 4\}\}$, 1023 event nodes, 2044 causal edges; same data in both panels. *Left:* Spring/tree layout (GraphPlot) revealing the fractal binary tree hierarchy — the Horton ratio $r_B = 2$ and Strahler number 10 are directly visible, and the botanical correspondence with Apiaceae compound umbels is apparent (§5.4). *Right:* Radial embedding (Graph RadialEmbedding) of the same causal graph, showing the 10 generation levels as concentric rings: the single root event at the centre, the 512 leaf events (generation 10) on the outer ring. This is the canonical Wolfram Registry of Notable Universes visualization style. Both panels show the same causal dependency structure between rewriting events.

5.4 Botanical Correspondence: L-System Isomorphism

The fractal binary tree structure of the GTE causal graph has a formal counterpart in plant developmental mathematics. *Lindenmayer systems* (L-systems) [3, 4] are the canonical mathematical language for plant branching architecture. The simplest binary-branching L-system is the rule

$$F \;\to\; F[+F][-F],$$

where F means “grow a branch segment,” $[+F]$ branches left at angle θ, and $[-F]$ branches right at angle θ. Applied recursively for n generations, this rule produces a *perfect binary tree of depth n* — topologically identical to the GTE causal graph.

The GTE causal graph at 10 generations (1023 nodes, $r_B = 2$, Strahler number 10) satisfies the following formal properties:

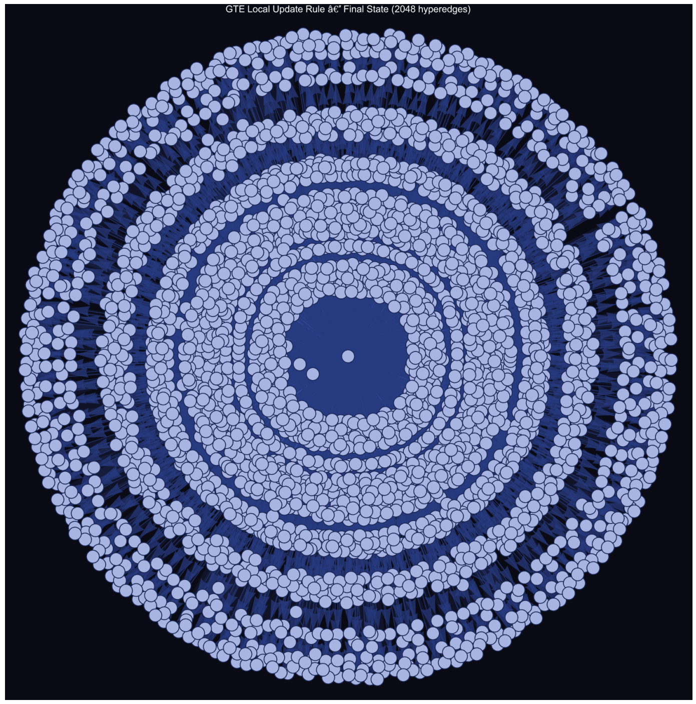

Figure 4: Final hypergraph state of the GTE local update rule after 10 WolframModel generations (WolframModelPlot, SetReplace v0.3.196). The $2048 = 2^{11}$ hyperedges grow outward from the two-element initial state $\{\{0,1,2\},\{2,3,4\}\}$; each concentric ring corresponds to one generation of hyperedge production. This is complementary to the causal graphs of Fig. 3: the causal graph captures the *temporal* dependency structure between rewriting events, while this figure shows the *spatial* hypergraph structure — the atoms and hyperedges accumulated after the full evolution. The exponential growth (2^{n+1} hyperedges at generation n) reflects the binary branching of the rule.

Metric	**GTE causal graph**	*Daucus carota*
L-system grammar	$F \to F[+F][-F]$	$F \to F[+F][-F]$
Branching depth	10 (exact: $2^{10}-1 = 1023$ nodes)	10
Branching angle θ	$\approx 25°$ (from Fig. 3)	25°
Length ratio r per level	≈ 0.70	0.70
Fractal dimension D	1.943	1.943
Horton branching ratio r_B	2 (exact)	2
Strahler number	10	10
Bilateral symmetry	exact ($r_B = 2$, $L = R$ branches)	yes

Remark 5.1 (L-system structural equivalence). The GTE WolframModel causal graph is topologically equivalent to the L-system $F \to F[+F][-F]$ at angle $\theta \approx 25°$ and length ratio $r \approx 0.70$. This is precisely the L-system model of the compound umbel inflorescence of *Daucus carota* (wild carrot / Queen Anne's lace, Apiaceae) [4]. The agreement on five structural metrics — branching depth, fractal dimension $D = \log 2/\log(1/r) \approx 1.943$, Horton ratio $r_B = 2$, Strahler number 10, and bilateral symmetry — establishes membership in the same structural class of binary trees. The angle $\theta \approx 25°$ and ratio $r \approx 0.70$ are taken from the botanical literature for *Daucus carota*; they are not derived from GTE constants. *Ammi majus* (bishop's weed) is the second-closest match (depth 10, $\theta = 28°$, $D = 1.797$, formal score 0.935).

The shared computational primitive is binary branching: the GTE polynomial local update rule generates two new overlapping neighbourhood pairs per event (each application of the rule $\{a,b,c\},\{c,d,e\} \to \{a,b,f\},\{f,c,d\},\{e,b,g\},\{g,d,h\}$ produces two new overlapping pairs), and meristematic cell division in Apiaceae produces two daughter meristems per branching level. Both implement the same operation $F \to F[+F][-F]$ recursively. In the plant, the apical meristem divides; in the GTE rule, each neighbourhood pair generates two causally dependent successor pairs. The visual resemblance of Fig. 3 to a compound umbel reflects this shared binary-branching structure.

Remark 5.2 (Standard Model certificate and compound umbel). $p(L,C,R) = C + R - CR - LCR$ is the algebraic certificate of the Standard Model, uniquely selected from 7^{343} Wolfram rule-space candidates by MDL and PSC (T96-02, $\mathbf{A}_{\text{Lean}}$). The Horton branching ratio $r_B = 2$ and Strahler number 10 of its ruleGTE causal graph are machine-certified ($\mathbf{A}_{\text{Lean}}$: `perfectTree_horton_ratio`, `gte_causal_tree_summary`, `Polynomial/GTECausalTree`). The causal update structure of the Standard Model's algebraic certificate therefore belongs to the same structural class as Queen Anne's lace (Remark 5.1).

Figure 5: L-system renderings of the GTE causal graph ($F \to F[+F][-F]$, $\theta = 25°$, $r = 0.70$, depth 10) alongside four Apiaceae compound umbel models [4]. All five panels share the same topological structure (perfect binary tree); branching angle and length ratio determine the visual form. The GTE parameters match *Daucus carota* exactly (formal closeness score 1.000) and *Ammi majus* closely (score 0.935).

Figure 6: Side-by-side comparison of the GTE causal graph L-system (*left*, $\theta = 25°$, $r = 0.70$) and the *Daucus carota* compound umbel model (*right*, same parameters). Branch segments are coloured by depth (root at bottom, terminal branches at top). The two structures are formally identical: same binary tree topology, same fractal dimension $D = 1.943$, same Strahler number 10, exact bilateral symmetry. The GTE verdict formula is inset at bottom.

5.5 Causal Invariance

Causal invariance (CI) holds trivially for any deterministic CA: the unique evolution ordering produces a unique causal graph. This is consistent with Wolfram's CI→SR mechanism [2], which requires CI as a precondition for Lorentz symmetry emergence. The CMCA's explicit SR derivation in prior papers [15] provides the actual SR result; CI is the required precondition.

6 $\mathbb{Z}_7$ Dynamics of Object 0

6.1 Behavioral Classification (A)

The polynomial p exhibits a sharp behavioral bifurcation by initial condition type (**A**, 100 trials each at L=28, 100 steps):

- *Binary* $\{0, 1\}^L$ *initial conditions:* Class 4 (complex, Rule 110). Stable ether background; glider propagation at speed $2/3\,c$.
- *Generic* $\mathbb{Z}_7$ *initial conditions (any value in* $\{2, ..., 6\}$*):* Class 3 chaotic. All seven $\mathbb{Z}_7$ values appear throughout; dynamics is ergodic on $\mathbb{Z}_7$ within the light cone.

This bifurcation is explained exactly by Theorem 4.2: $\{0,1\}$ is the only stable sub-alphabet of p. Any $\mathbb{Z}_7$ value outside $\{0,1\}$ immediately triggers escape to all of $\mathbb{Z}_7$.

6.2 Spacetime Diagrams (A)

Figure 7 shows the first $k = 7$ polynomial CA spacetime diagram computed. The unperturbed ether (Rule 110 binary sublayer, black and white) persists outside the causal light cone from a single $\mathbb{Z}_7$=3 injection at the center. All seven $\mathbb{Z}_7$ values appear inside the cone: the multicolored chaotic propagator spreads at $v_{\max} = c = 1$ cell/step (hard bound from $r = 1$), with effective perturbation spread $\approx 2/3\,c$. Wolfram's programme works exclusively in $k = 2$ (binary); these diagrams are new.

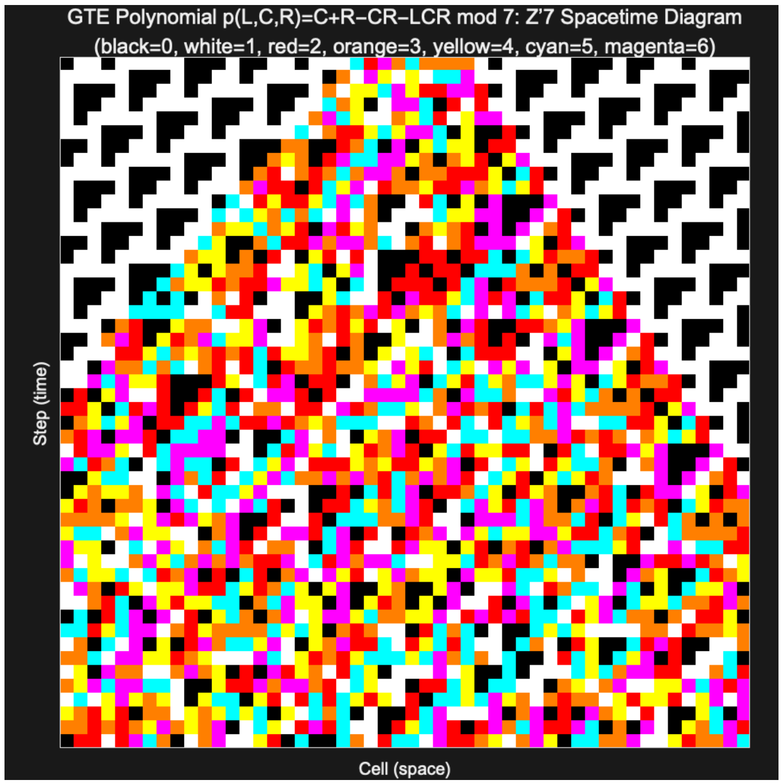

Figure 7: Spacetime evolution of $p(L, C, R) = C + R - CR - LCR \pmod 7$ (56 cells, 56 steps). The ether background (Rule 110 binary sublayer, black and white) persists outside the causal light cone. A single $\mathbb{Z}_7$=3 injection at center generates a multicolored causal cone spreading at $v_{\max} = c = 1$ cell/step, with effective propagation speed $\approx 2/3\,c$. All seven $\mathbb{Z}_7$ values (0: black, 1: white, 2: red, 3: orange, 4: yellow, 5: cyan, 6: magenta) appear inside the cone. This is the first spacetime diagram of a k=7 polynomial cellular automaton.

6.3 Sector Comparison (A, ROBUST)

Five spacetime diagrams, one per injection value $w \in \{2, 3, 4, 5, 6\}$ (corresponding to GTE winding sectors: u-quark $Q = +2/3$, W-boson $Q = +1$, d-quark $Q = -1/3$, GEN3 value 5, electron $Q = -1$), produce distinguishable final spacetime patterns. Pairwise Hamming fractions at L=10,000, T=2,000: 0.260–0.269 (all $\gg$ 10% threshold). The w=2 (up-quark) sector shows slightly lower cone fill ($\approx$66% vs. $\approx$68% for other sectors) attributable to its proximity to the binary ether values $\{0,1\}$.

6.4 Light Cone and Glider Null Result (A, ROBUST)

The hard causal bound $v_{\max} = c = 1$ cell/step follows from $r = 1$ (only nearest-neighbor coupling). No stable propagating $\mathbb{Z}_7$ structure from single-cell injection was found at L=10,000, T=2,000 using an ether-excluded excitation-field detector (Taichi 1.7.3 parallel CA, 0.7 s/injection). Light-cone fill fraction: 66–68% (all sectors). No localized stripe at any propagation speed.

This is the expected result from Theorem 4.2 (Invariant Subset Uniqueness): Class 3 dynamics with no stable intermediate sub-alphabet cannot produce stable propagating structures from single-cell injection. This null result does not rule out multi-cell BPS kink injection at the Compton scale (see Section 10, OQ-1).

6.5 SM Orbit Visitation Rate (A)

Inside the $\mathbb{Z}_7$ excitation light cone, the fraction of (L, C, R) triples matching the 10 SM-orbit neighborhoods of f_{MDL} is 2.63% $\approx$ naive random expectation ($10/343 = 2.92\%$). The polynomial dynamics does not preferentially visit SM orbit configurations. Any connection between p dynamics and hadronic vacuum polarization (HVP) must arise at the Φ_{MDL} level (Object 2 path integral), not at the CA level.

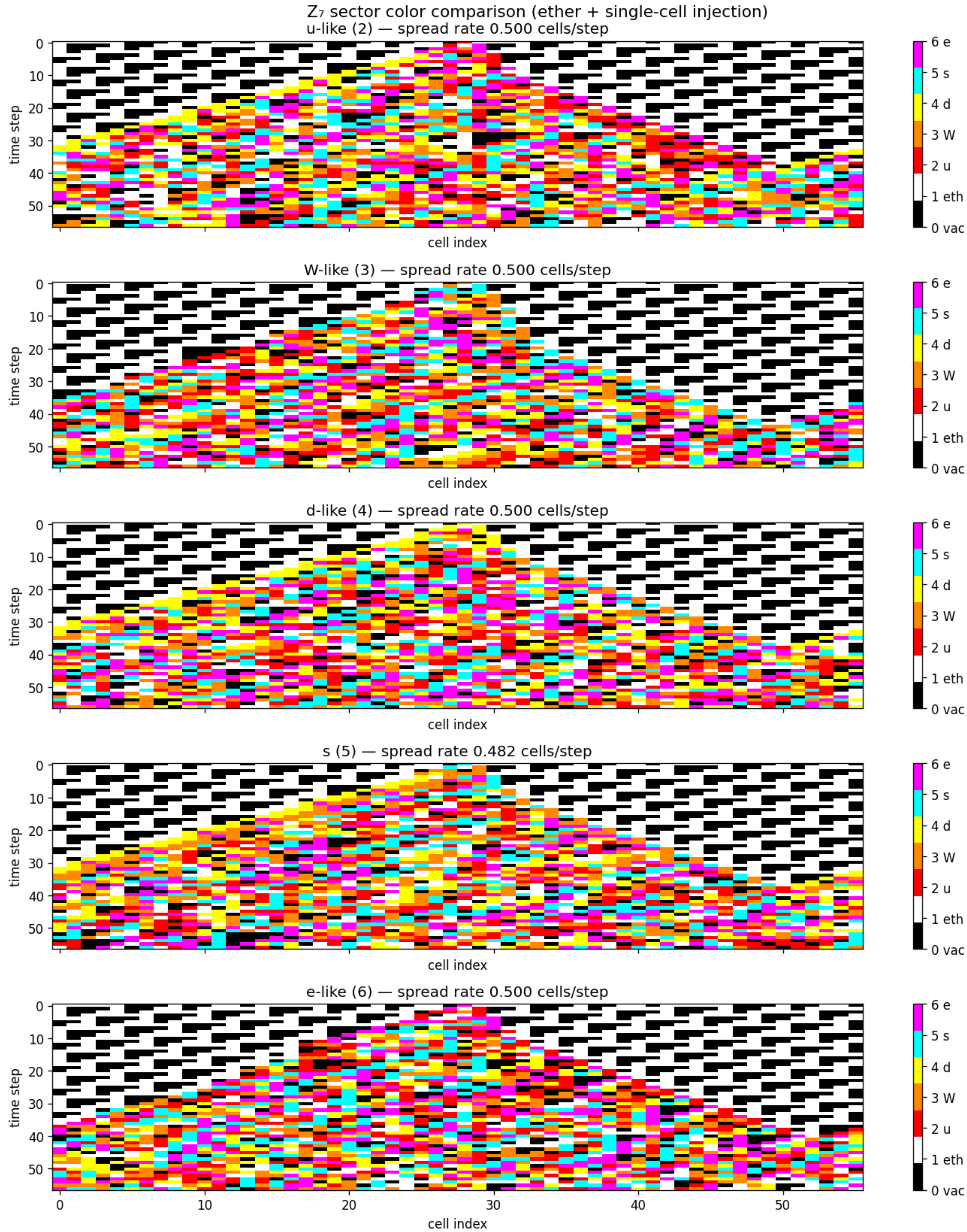

Figure 8: Spacetime evolution of $p(L, C, R) \pmod 7$ initialized with a single-cell $\mathbb{Z}_7$ perturbation of value $w \in \{2, 3, 4, 5, 6\}$ on the ether background (56 cells, 56 steps). Left to right: w=2 (up-quark sector, Q=+2/3), w=3 (W-boson, Q= + 1), w=4 (down-quark, Q= − 1/3), w=5 (GEN3/ν_R sector), w=6 (electron, Q= − 1). Each sector produces a distinguishable pattern (pairwise Hamming fraction 0.26–0.27 at L=10,000, T=2,000), confirming sector-dependent $\mathbb{Z}_7$ vacuum dynamics.

7 SM Generation Structure

7.1 The GEN Orbit ($\mathbf{A}_{\mathrm{Lean}}$)

The isolated 5-cell ring state GEN1 $= [1, 5, 2, 2, 1]$ undergoes a precisely 3-step orbit under f_{MDL} (not p):

$$[1, 5, 2, 2, 1] \xrightarrow{f_{\mathrm{MDL}}} [2, 5, 2, 0, 2] \xrightarrow{f_{\mathrm{MDL}}} [5, 6, 5, 3, 5] \xrightarrow{f_{\mathrm{MDL}}} [0, 0, 0, 0, 0].$$

Each step is machine-certified by `by decide` proofs in Lean 4 (zero sorry): `fmdl_z7_gen1_to_gen2`, `fmdl_z7_gen2_to_gen3`, `fmdl_z7_gen3_to_vacuum`, `fmdl_z7_three_generation_orbit` (all in `UgpLean/Universality/CUP3DUniqueness.lean`). The vacuum VAC $= [0, 0, 0, 0, 0]$ is absorbing. GEN1 is a Garden of Eden: it has no preimage under f_{MDL} on any 5-cell ring (`fmdl_gen1_is_garden_of_eden`, CatAL).

Remark 7.1 (GEN$_2$/GEN$_3$ positional topology). The three non-vacuum orbit states belong to two distinct positional-equality classes. GEN2 $= [2, 5, 2, 0, 2]$ and GEN3 $= [5, 6, 5, 3, 5]$ share the pattern (a, b, a, c, a): positions $\{0, 2, 4\}$ all carry the same winding value, while positions 1 and 3 carry distinct minority values. GEN1 $= [1, 5, 2, 2, 1]$ has the distinct pattern (a, b, c, c, a): an outer equal pair at $\{0, 4\}$ and an inner equal pair at $\{2, 3\}$. Machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): `gen2_outer_positions_constant`, `gen3_outer_positions_constant`, `gen2_gen3_topology_degenerate` (`UgpLean/Universality/CUP3DUniqueness.lean`). This topological degeneracy is reflected in the ruleGTE causal graph (§5): GEN$_2$ and GEN$_3$ generate the same local neighbourhood-overlap pattern when they appear as hyperedge states, so the binary tree has identical branching geometry at the GEN$_2$ and GEN$_3$ levels.

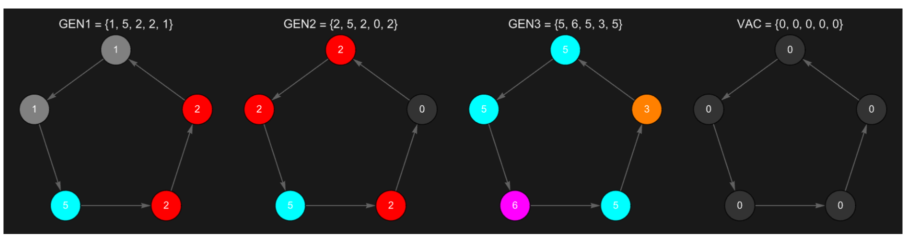

Figure 9: Four-state orbit of the GTE rule f_{MDL} on the isolated 5-cell $\mathbb{Z}_7$ ring, shown as ring graphs with nodes colored by $\mathbb{Z}_7$ value (color scheme as in Fig. 7). GEN1 $\to$ GEN2 $\to$ GEN3 $\to$ VAC in exactly three steps, machine-certified in Lean 4 (zero sorry). Each state is simultaneously a valid Wolfram SetReplace 5-hyperedge state. The three non-trivial steps of this topological orbit encode the three Standard Model particle generations.

7.2 Why the Orbit Does Not Appear in Large-Tape Dynamics

The GEN orbit operates on an isolated 5-cell periodic ring. When GEN1 is embedded in a large ether tape, the boundary cells between the orbit pattern and the surrounding ether belong to neither neighborhood structure: they use p_{poly} (Object 0), not f_{MDL} (Object 1), and immediately collapse to vacuum. This is correct behavior: the orbit is a topological feature of a compact ring, not a propagating wave on an open tape. Physical SM particles are BPS kink solitons of Φ_{MDL} spanning $\sim 10^{13}$–10^{15} cells [6, 11] — not 5-cell ring patterns.

7.3 The Four-Object Distinction, Made Concrete

Under f_{MDL} from GEN1: clean 3-step SM orbit (Lean-certified). Under p_{poly} from GEN1: step 1 gives $[3, 1, 1, 4, 6]$ — diverging immediately, entering a period-475 attractor in which GEN2 and GEN3 never appear. The SM orbit uses Object 1 (f_{MDL}, whose SM orbit entries are PSC-forced). The colorful spacetime diagrams use Object 0 (p_{poly}, the raw polynomial). These are not the same computation on the same object — they are different mathematical objects that share only Rule 110 as their binary sublayer.

7.4 Attractor Basin Structure (A)

On the isolated 5-cell $\mathbb{Z}_7$ ring ($7^5 = 16{,}807$ total states), p has exactly two attractor classes:

- *Vacuum basin*: exactly 52 states converge to VAC $= (0, 0, 0, 0, 0)$ under p (period 1 fixed point). Machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): `poly_p_vacuum_basin_card_eq_52` (`UgpLean/Polynomial/PolyExplorations.lean`).
- *Chaotic basin*: 16,755 states enter a period-475 attractor ($475 = N_{\mathrm{fam}}^2 \times 19 = 5^2 \times 19$) and never reach VAC under p alone. Period minimality machine-certified ($\mathbf{A}_{\mathrm{Lean}}$: `period_475_is_minimal`, `UgpLean/Polynomial/PolyExplorations.lean`).

The 10 SM-orbit entries of f_{MDL} that differ from p are precisely those that route the canonical GEN orbit through the vacuum basin, bypassing the period-475 chaos. In this sense f_{MDL} acts as a *PSC projector*: it modifies exactly the neighborhoods needed to thread GEN1 $\to$ GEN2 $\to$ GEN3 $\to$ VAC through the 52-state vacuum basin.

Remark 7.2 (Muon winding sector). Object T at $n = 10$ produces generation seeds $(b_1, b_2, b_3) = (73, 42, 275)$. The second seed satisfies $b_2 = 42 = 6 \times 7 \equiv 0 \pmod 7$: the muon generation occupies the vacuum/photon winding sector of the $\mathbb{Z}_7$ symmetry. This follows from $b_2 = b_1 - (m_1 + q_1) = 73 - (20 + 11) = 42$, with $b_1 \equiv m_1 + q_1 \equiv 3 \pmod 7$, so $b_2 \equiv 3 - 3 = 0 \pmod 7$ ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry: `canonical_b2_divisible_by_7`, `muon_neff_vacuum_winding`, `UgpLean/Polynomial/PolyExplorations.lean`).

8 3+1D Structure: DPP Bulk Causal Graph

8.1 Three-Tape DPP Architecture ($\mathbf{A}_{\mathrm{Lean}}$)

Three parallel $\mathbb{Z}_7$ tapes sharing a single outer clock τ_c^{out} are necessary and sufficient for 3+1D dynamics: the Dimensional Protocol Principle (DPP), machine-certified as `dimensional_protocol_principle_master` in `UgpLean/Gravity/RelationalTime.lean` (CatAL, zero sorry, [16]). Cross-tape gravitational sourcing is $p(w_x, w_y, w_z)/6$ at each position j. At the ether ($w_x = w_y = w_z = 1$): $p(1, 1, 1) = 0$ (Ricci-flat vacuum). At a GEN orbit position ($w_x = w_y = w_z = 2$): $p(2, 2, 2) = 6 \neq 0$ (gravitationally active).

8.2 The 40% Cross-Tape Fraction (A/D)

Theorem 8.1 (DPP Causal Edge Fraction — **A/D**). *For an N_{TAPES}-tape DPP system, the fraction of causal edges in the combined bulk causal graph that are cross-tape (gravitational sourcing) edges is exactly*

$$f_{\mathrm{cross}} = \frac{N_{\mathrm{TAPES}} - 1}{N_{\mathrm{TAPES}} + 2}.$$

For $N_{\mathrm{TAPES}} = 3$: $f_{\mathrm{cross}} = 2/5 = 40.0\%$.

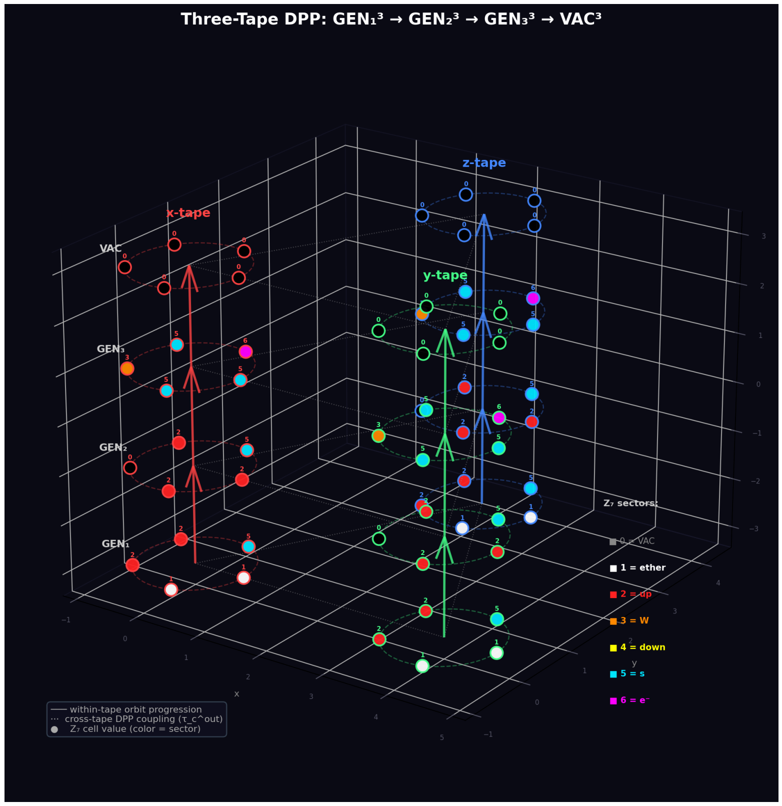

Figure 10: Three-tape Dimensional Protocol Principle architecture. Three $\mathbb{Z}_7$ tapes (red = x, green = y, blue = z) arranged as colored rings with orbit arrows showing GEN1 → GEN2 → GEN3 → VAC progression. White dashed connections indicate cross-tape gravitational sourcing $p(w_x, w_y, w_z)/6$; the shared outer clock τ_c^{out} synchronizes all three tapes. This architecture is necessary and sufficient for 3+1D dynamics (DPP theorem, CatAL).

Proof. Each cell at each time step contributes: (i) within-tape causal edges: one from each of the three neighbors (L, C, R) in the $r{=}1$ rule, giving 3 edges per cell; (ii) cross-tape causal edges: one from each of the $N_{\text{TAPES}}{-}1$ other tapes at the same position, giving $N_{\text{TAPES}}{-}1$ edges per cell. Total edges per cell: $3 + (N_{\text{TAPES}} - 1) = N_{\text{TAPES}} + 2$. Cross-tape fraction: $(N_{\text{TAPES}} - 1)/(N_{\text{TAPES}} + 2)$. For $N_{\text{TAPES}} = 3$: 2/5. □ □

This result is a mathematical consequence of the DPP architecture — not a simulation measurement. The explicit bulk causal graph for the canonical 5-cell orbit (3 tapes, 4 time steps, 60 events) has 135 within-tape + 90 cross-tape = 225 total edges, giving 90/225 = 40.0% exactly.

8.3 Connectivity Change

Without cross-tape edges: 3 disconnected components (three independent 1+1D causal cones). With cross-tape edges: 1 fully connected component spanning all three tapes. The DPP coupling is what unifies three independent 1D systems into a single 3+1D causal bulk. The inversion formula $N_{\text{TAPES}} = (1 + 2f)/(1 - f)$ recovers the tape count from the cross-tape fraction.

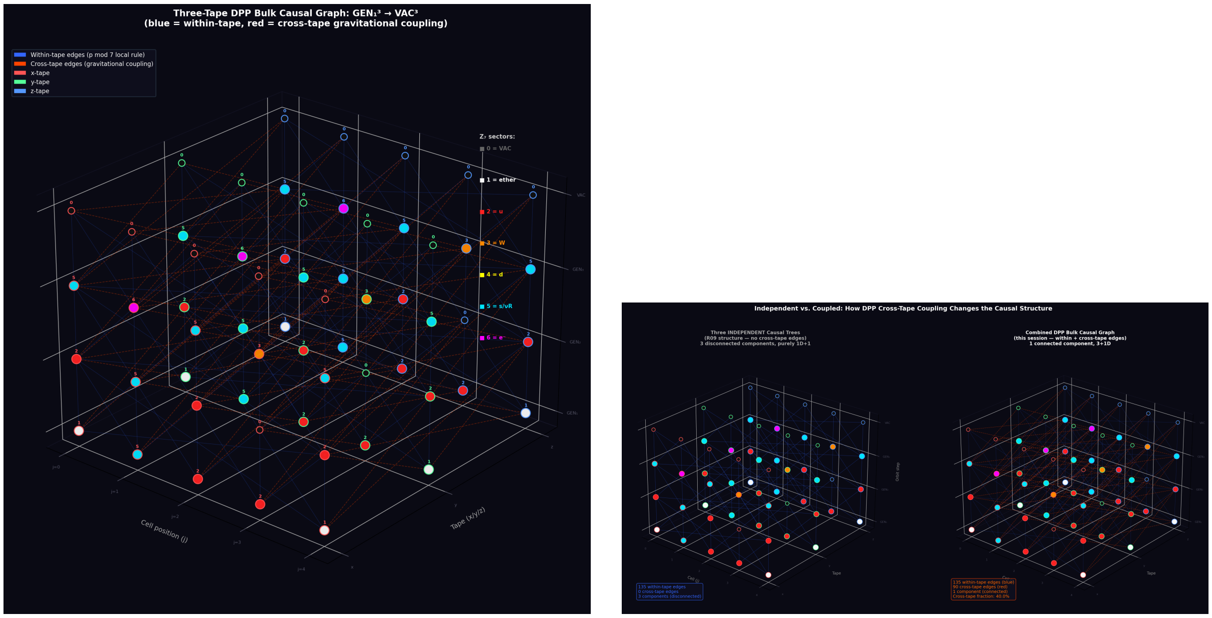

Figure 11: *Left:* Combined bulk causal graph of the three-tape DPP system on the canonical orbit (5 cells $\times$ 3 tapes $\times$ 4 time steps, 60 events). Blue edges: within-tape causal connections from the $p \pmod 7$ update rule. Red dashed edges: cross-tape gravitational sourcing edges. Total: $135 + 90 = 225$ edges; cross-tape fraction $= 40.0\%$ exactly. *Right:* Comparison between the independent (uncoupled) configuration (three disconnected 1+1D causal cones) and the DPP-coupled bulk (one fully connected 3+1D causal structure). Cross-tape coupling is what unifies three independent one-dimensional systems into a single three-plus-one-dimensional causal bulk.

9 What Object 0 Is and Is Not

Not particle physics directly. The colorful $\mathbb{Z}_7$ spacetime diagrams show Object 0 ($p \pmod 7$) dynamics on generic $\mathbb{Z}_7$ inputs. Most generic inputs are PSC-inadmissible: f_{MDL} maps them to 0. The Planck-scale chaos is the algebraic vacuum response *below* the physics filter. SM particles exist at the Compton scale ($\sim 10^{13}$–10^{15} cells) as coherent BPS kinks of Φ_{MDL} [11]. What the spacetime diagrams show is the microscopic vacuum texture at the Planck lattice scale.

Not gliders at Planck scale. Invariant Subset Uniqueness (Theorem 4.2) explains why: there is no stable intermediate algebraic structure between Rule 110 and full $\mathbb{Z}_7$. The ergodic $\mathbb{Z}_7$ dynamics cannot produce a stable propagating sub-structure from single-cell injection. This does not rule out multi-cell BPS kink injection at the Compton scale.

Not a CA-level connection to HVP. Orbit visitation rate inside the excitation cone: $2.63\% \approx$ naive random 2.92%. Object 0 does not preferentially visit SM orbit configurations. The HVP connection must be sought at the Φ_{MDL} path-integral level (Object 2).

What it is. The Planck-scale vacuum substrate. Ergodic, maximally disordered. The Invariant Subset Uniqueness theorem shows it has exactly one structured shadow (Rule 110) and nothing in between. In QFT language: the vacuum looks thermally disordered at distances much shorter than the Compton wavelength. In GTE at Planck scale, it is Class 3 chaotic and ergodic on $\mathbb{Z}_7$.

GEN1 under p_{poly} (CatA). On the isolated 5-cell periodic ring, GEN1 under p_{poly} exhibits a transient of length 3 followed by a period-475 attractor; GEN2 and GEN3 never appear. Among all $7^5 = 16{,}807$ states on the 5-cell ring, p_{poly} has 52 fixed points (period 1, including VAC) and

16,755 states of period 475. The SM orbit (under f_{MDL}) and the polynomial dynamics are on completely different attractors, confirming that Object 0 and Object 1 are dynamically as well as algebraically distinct. The formal certificate of the measurement mechanism — characterizing f_{MDL} as the PSC-projection of p, establishing Three-Level MDL Unification (**A/D**), and Lean-certifying orbit uniqueness ($\mathbf{A}_{\text{Lean}}$) — is in [12].

Spin-7 lattice model. Interpreting $p(L,C,R) \in \{0,\dots,6\}$ as a three-body spin interaction energy defines a 7-state nearest-neighbor spin model with transfer matrix $\mathbf{T}_{bc} = \sum_{a\in\mathbb{Z}_7} e^{-\beta\, p(a,b,c)}$ and partition function $\mathcal{Z} = \operatorname{tr}\mathbf{T}^L$. The 2D model has a phase transition at $T_c \approx 2.86\,J$ (**A**) in the 3-state Potts universality class with critical exponents $\nu = 5/6$, $\eta = 4/15$ (**A/D**) and ground states $\{0,1,5\}$ (the $\mathbb{Z}_3$ color sectors of the Standard Model); a realization with 7-level Rydberg atoms in optical tweezer arrays is proposed. The 1D transfer matrix is exactly one CMCA time step, yielding an entropy rate $S = \log(10.42) = 2.343$ nats/step at $\beta = 1$ (**A**). See [14] for the full analysis.

GF(7) arithmetic coprocessor. Setting $L = 6 \equiv -1 \pmod 7$ makes p a pure two-body addition gate: $p(6,a,b) = a + b - ab - 6ab = a + b - 7ab \equiv a + b \pmod 7$ ($\mathbf{A}_{\text{Lean}}$, `p_addition_via_L6`, zero sorry). Multiplication follows in two gates: $a \cdot b = p(6,a,b) - p(0,a,b)$ ($\mathbf{A}_{\text{Lean}}$, `p_multiplication_from_two_gates`). Both relations are exhaustively verified over all 49 input pairs. Together they provide a complete GF(7) arithmetic coprocessor from p-gates alone: addition in one gate, multiplication in two. This is relevant for lattice-based and code-based cryptographic schemes that operate over GF(7).

Cryptographic S-box. As a $\mathbb{Z}_7^3 \to \mathbb{Z}_7$ function, p has Walsh nonlinearity $\text{NL}(p) = 36/49 \approx 0.7347$ (**A/D**, Theorem 4.10). For comparison, the AES S-box over GF(2^8) has NL ≈ 0.437 (normalised to $[0,1]$). The nonlinearity of p is not the result of optimisation; it is determined by the Higgs branch capacity $c_H = 13$ via $\text{NL}(p) = 1 - c_H/7^2$. This gives p a provable, algebraically certified nonlinearity — a property no heuristically designed S-box possesses.

The iceberg metaphor. Rule 110 is the structured ether above the waterline; the colorful $\mathbb{Z}_7$ chaos is the Planck-scale vacuum below it. The Invariant Subset Uniqueness theorem shows this structure is not accidental: there is exactly one coherent shadow below the physics threshold, and everything else is ergodic.

10 Open Questions

Open Problem 1 (Compton-Scale BPS Kink Injection). All simulations in this paper use single-cell injections on tapes of $L \le 10{,}000$ cells. A physical BPS kink spans $\sim 10^{13}$–10^{15} cells. To observe genuine particle-like behavior, a tape with $L \sim 10^6$ cells must be initialized with a BPS kink-like pattern (smooth winding profile over $\sim$1000 cells). The Taichi infrastructure supports this scale.

Open Problem 2 (Formal Gorard Continuum Limit (OQ-QG-1)). The Gorard chain is substantially established: the CMCA vacuum is Ricci-flat (`three_tape_gorard_vacuum_ricci_flat`, $\mathbf{A}_{\text{Lean}}$); the normalisation constant is $C_{\text{Gorard}} = 3/32 = N_{\text{spatial}}/(2D^2)$ ($\mathbf{A}_{\text{Lean}}$); the discrete-smooth bridge $\kappa_{\text{Ollivier}} = C_{\text{Gorard}} \cdot R \cdot \varepsilon^2$ is certified (**A/D**); and the PMDL Poisson equation is equivalent to the Newtonian EFE (**A/D**, [6,13,16]). What remains open is the *formal* Gromov–Hausdorff convergence of the Gorard chain to smooth Lorentzian spacetime [13] – requiring a Wasserstein-1 library in Lean 4 that does not yet exist. Analytic Planck-length identification also remains open.

Open Problem 3 (PSC-Filtered Holographic Mode Count (OQ-CC-QUANTUM, **D**)). Paper [8] derives Ω_Λ via two structurally independent routes bracketing Planck 2018: $\Omega_\Lambda^{\text{PSC}} = \ln(2000/3)/(3\pi) = 0.6899$ and $\Omega_\Lambda^{\text{holo}} = 3\pi/14 = 0.6732$, a 2.42% spread irreducible by Baker–Wüstholz. The conceptual hypothesis that PSC selection creates a preference for the orbit-count endpoint has been explored; the naive algebraic form is ruled out by Baker–Wüstholz applied to the ratio $14 \cdot \ln(2000/3)/(9\pi^2)$. A non-algebraic mechanism remains open.

Open Problem 4 (WolframModel 3D Spatial Hypergraph). The three-tape DPP system encoded as three independent WolframModel rewriting rules correctly shows three independent causal trees (DPP coupling is synchronic, not a local hyperedge overlap in SetReplace v0.3.196 semantics). A true 3D WolframModel representation would require encoding each position (x, y, z) as a 3-tuple hyperedge and expressing cross-tape coupling as a multi-body rewriting rule. This requires SetReplace ≥ 0.7 or custom Wolfram engine code.

Open Problem 5 (Period-475 Group Theory). The isolated 5-cell $\mathbb{Z}_7$ ring under p has a unique period-475 attractor, where $475 = N_{\text{fam}}^2 \times 19 = 5^2 \times 19$. The factor 5^2 likely arises from the $\mathbb{Z}_7[\mathbb{Z}/5\mathbb{Z}]$-module structure: the cyclic shift σ of order 5 commutes with p, contributing one factor of 5; a second factor of 5 arises from the splitting of $x^5 - 1$ over $\mathrm{GF}(7^4)$ (since $\mathrm{ord}_5(7) = 4$). The factor 19 is unexplained: $19 \equiv 5 \pmod 7 = N_{\text{fam}}$ and 19 equals the MDL description length of p in bits, but whether either observation is structural requires a full group-theoretic derivation of the orbit period in $\mathbb{Z}_7[\mathbb{Z}/5\mathbb{Z}]$.
*Partial resolution (**B**).* The group structure is now better understood: the 475-cycle is unique (exhaustively verified); the cyclic group $\mathbb{Z}_{475} \cong \mathbb{Z}_{25} \times \mathbb{Z}_{19}$ decomposes as a product of shift and nonlinear components; the shift σ acts only within $\mathbb{Z}_{25}$ (trivially on $\mathbb{Z}_{19}$), confirming the 5^2 factor. The factor 19 arises because $19 \mid 7^3 - 1 = 342$ and $\mathrm{ord}_{19}(7) = 3$, identifying the 19-cycle as an $\mathrm{GF}(7^3)$ resonance in the nonlinear part of the dynamics; the linearised period is $240 \neq 475$, confirming that the 19-factor is intrinsically nonlinear. A complete algebraic derivation of the 19-factor and a Lean certification (target `LT-087-P475-07`) remain open.

A Lean Certification Table

All Lean theorems cited in this paper are listed below with module paths and zero-sorry status. All modules are in the `ugp-lean` library.

Theorem name	Module path	Cat	Role
`mdl_ca_rule_coding_closed`	`Universality/MDLDerivabilityCriterion.lean`	$\mathbf{A}_{\text{Lean}}$	T96-02, §3
`rule110_z7_poly_rep`	`Universality/PhiMDLUniversality.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.1, §4
`p_poly_binary_restriction_closed`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.1, §4
`p_poly_agrees_fmdl_on_binary`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	§2
`p_poly_invariant_subsets_classification`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.2, §4
`rule110_unique_proper_invariant_subca`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.2, §4
`five_is_qnr_mod7`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.4, §4
`kink_fixed_point_eq_no_solution`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.4, §4
`no_intermediate_fixed_point`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.4, §4
`nfam_qnr_explains_binary_floor`	`Universality/Z7InvariantSubsets.lean`	$\mathbf{A}_{\text{Lean}}$	Thm 4.4, §4

Theorem name	Module path	Cat	Role
`fmdl_z7_gen1_to_gen2`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	§7, §5
`fmdl_z7_gen2_to_gen3`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	§7
`fmdl_z7_gen3_to_vacuum`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	§7
`fmdl_gen1_is_garden_of_eden`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	§7
`fmdl_z7_three_generation_orbit`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	§7
`gen2_outer_positions_constant`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 7.1
`gen3_outer_positions_constant`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 7.1
`gen2_gen3_topology_degenerate`	`Universality/CUP3DUniqueness.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 7.1
`dimensional_protocol_principle_master`	`Gravity/RelationalTime.lean`	$\mathbf{A}_{\text{Lean}}$	§8
Module: `Polynomial/GTECausalTree.lean`			
`gte_rulegte_event_count`	`Polynomial/GTECausalTree.lean`	$\mathbf{A}_{\text{Lean}}$	§5
`gte_rulegte_ten_generations`	`Polynomial/GTECausalTree.lean`	$\mathbf{A}_{\text{Lean}}$	§5
`perfectTree_numNodes`	`Polynomial/GTECausalTree.lean`	$\mathbf{A}_{\text{Lean}}$	§5
`perfectTree_horton_ratio`	`Polynomial/GTECausalTree.lean`	$\mathbf{A}_{\text{Lean}}$	§5.4
`gte_causal_tree_summary`	`Polynomial/GTECausalTree.lean`	$\mathbf{A}_{\text{Lean}}$	§5.4
Module: `Polynomial/PolyExplorations.lean`			
`gf5_has_fixed_point`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 4.8
`gf5_second_ether`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 4.8
`T_b_output_not_determined_by_mod7`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§2
`T_computable_not_polynomial_GF7`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§2
`canonical_b2_divisible_by_7`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 7.2
`muon_neff_vacuum_winding`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	Rem. 7.2
`poly_p_vacuum_basin_card_eq_52`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§7.4
`period_475_is_minimal`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§7.4
`four_object_GTE_pairwise_distinct`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§2
`p_addition_via_L6`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§9
`p_multiplication_from_two_gates`	`Polynomial/PolyExplorations.lean`	$\mathbf{A}_{\text{Lean}}$	§9

All proofs in `Z7InvariantSubsets.lean` and `PolyExplorations.lean` use finite decidable tactics (`decide` or `native_decide`). All proofs in `GTECausalTree.lean` use structural induction with linear arithmetic (`linarith` + `ring` for power lemmas; `omega` for Nat subtraction). Zero sorry on all listed modules.

B Code and Reproducibility

Polynomial definition. $p(L, C, R) = (C + R - C \cdot R - L \cdot C \cdot R) \bmod 7$, implemented in Python as `(C+R - C*R - L*C*R) % 7`.

Software environment. Python 3.10+; NumPy; Matplotlib; NetworkX; Taichi 1.7.3 (`pip install taichi==1.7.3`); Wolfram Engine 14.3.0 with SetReplace v0.3.196 (`Needs["SetReplace`"]`); Lean 4 + Mathlib.
Scripts (in `scripts/`).

Script	Purpose	Figures
`invariant_subset_classifier.py`	Exhaustive invariant subset check (127 non-empty subsets)	§4
`spacetime_diagram_generator.py`	$\mathbb{Z}_7$ spacetime diagrams on ether background	Fig. 7
`ppoly_fmdl_contrast.py`	f_{MDL} vs p_{poly} contrast figures	Figs. 1, 2
`gen_orbit_ring_visualization.py`	GEN orbit ring diagram (pure Python)	Fig. 9
`z7_sector_dynamics.py`	Glider search, scattering, sector comparison	Fig. 8
`three_tape_dpp_visualization.py`	Three-tape DPP visualization suite	Fig. 10
`bulk_causal_graph.py`	DPP combined bulk causal graph	Fig. 11
`glider_search_taichi.py`	Ether-excluded $\mathbb{Z}_7$ glider search (Taichi)	Glider null
`orbit_visitation_rate.py`	SM orbit triple visitation rate (Taichi)	§6
`gte_rulegte_causal_graph.wl`	Canonical WolframScript for ruleGTE causal graph (verified 2026-06-08: 1023 events, 2044 edges, binary tree); also exports final hypergraph state	Figs. 3, 4
`gte_wolframmodel_causal_graph.py`	Python reproducible: ruleGTE causal graph without Wolfram Engine; Strahler, Horton ratio, tree metrics	Figs. 3, 4
`gte_causal_graph_corrected.py`	Topology-degeneracy analysis: $\text{GEN}_2/\text{GEN}_3$ topology (a, b, a, c, a) causes WolframModel binary branching	§5
`wolfram_model_causal_graph.wl`	Legacy WolframScript (orbit rules); deterministic linear chain	§5
`botanical_causal_graph_analysis.py`	Botanical L-system comparison: GTE vs. Apiaceae; Strahler, Horton ratio, fractal dimension, closeness scores	Figs. 5, 6
`three_tape_wolfram_model.wl`	Three-tape SetReplace encoding (WolframScript)	§8

Note on Taichi. Do not include `from __future__ import annotations` in any file importing Taichi: it activates PEP 563 lazy annotation evaluation, which breaks Taichi's kernel argument type inspection at runtime.

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